

The Ultimate Space-Time Travel Planner

Project Description

This is a text-based Python program designed for 8th-grade computer science students to simulate space travel. The program helps users plan rocket trips to celestial destinations across our solar system. By combining the Python math library for planetary gravity adjustments and the datetime library for calendar shifts, it calculates travel duration, exact future landing dates, and biological metric changes (weight and age) on other planets.

Features

- **Dynamic Date Tracking:** Uses the datetime library to process departure dates and calculate exact calendar arrival dates.
- **Planetary Physics Modeling:** Integrates the math library to perform floor rounding for cosmic weight changes based on planetary gravity.
- **Error and Input Validation:** Handles invalid keystrokes and wrong date patterns securely using standard error-catching loops.
- **Cosmic Conversion Factors:** Features automated code paths to map precise orbital year lengths for your age across different planetary horizons.

How to Run the Program

1. Ensure that you have Python 3 installed on your computer.
2. Download or save the code file as `space_travel_planner.py`.
3. Open your computer's terminal, command prompt, or development workspace (such as Thonny or VS Code).
4. Run the program using the command: `python space_travel_planner.py` or press F5 inside your IDE.
5. Follow the on-screen instructions to select planets and calculate your space metrics.

Example Output

```
=====
THE ULTIMATE SPACE-TIME TRAVEL PLANNER
=====
Enter your weight on Earth (lbs): 120
Enter your current age (years): 14
Enter departure date (YYYY-MM-DD) [Leave blank for today]: 2026-08-28
Select your celestial destination:
1. Mars (The Red Planet)
2. Jupiter (The Gas Giant)
3. Venus (The Evening Star)
Enter choice (1-3):
1 Calculating flight trajectory... Done!
----- TRIP SUMMARY -----
Destination: Mars Departure Date: Friday, August 28, 2026
Estimated Flight Time: 5,088 hours (212 days)
Estimated Arrival Date: Sunday, March 28, 2027
----- PLANETARY METRICS -----
Your Weight on Mars: 45.0 lbs (38.0% of Earth gravity)
Your Age on Mars: 7.4 Mars years
=====
```

Contributors

Member 1: Lyan Ezekiel P. Tolentino- Made the first version of README and CHANGELOG and enhanced planetary metrics

Member 2: Ronand Jeth E. Macapanas- Contributed in adding new features (Planets) and making the initial baseline of the program.

Member 3: Jaden Lawrence O. Yusiong - Contributed in adding new features (Distance and Datetime Tracking)